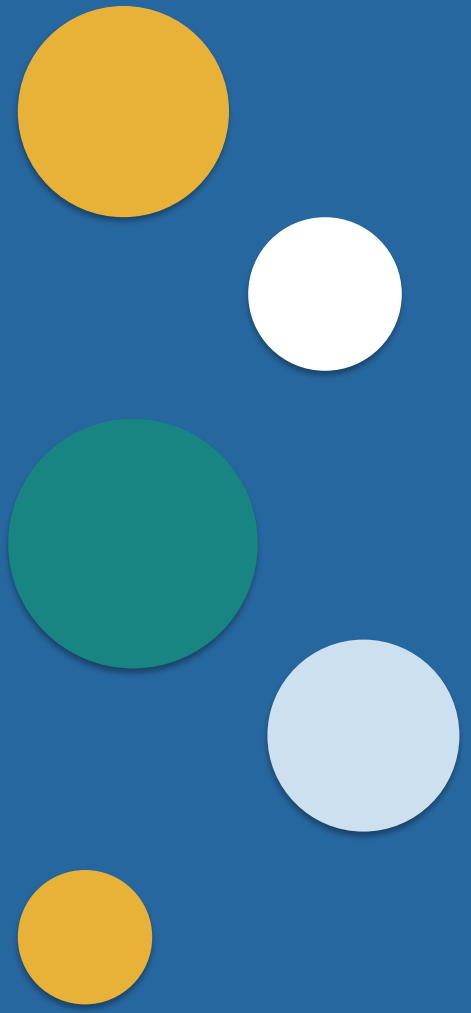
ACADEMIC PROJECT PRESENTATION

Working with Text in Python

Integrating Ethical Principles with the Ideals and Objectives of the United Nations

Project Review • Python • Text Analytics • Responsible Data Use

ETHICS • DATA • HUMANITY



Project Overview

Purpose, scope, and central question

What is studied?

Text from public UN reports, policy documents, statements, and SDG materials.

What is applied?

Python string methods, text cleaning, keyword searches, word counts, and introductory NLP.

What guides the work?

Human rights, fairness, privacy, transparency, inclusion, and accountability.

Central question

How can Python help organize and analyze UN-related text while ensuring that the methods and interpretations remain ethical and responsible?

Project Objectives

Technical learning and ethical analysis

- Apply Python string operations to textual information.
- Clean and standardize text for consistent analysis.
- Identify recurring words, phrases, and broad themes.
- Connect text-analysis topics with relevant UN Sustainable Development Goals.
- Evaluate risks involving bias, privacy, context, and misinterpretation.
- Document sources, methods, assumptions, and limitations.

Methodology and Workflow

A reproducible pipeline from source documents to interpretation

1. Collect

Use documented public UN sources

→

2. Prepare

Inspect data and handle duplicates

→

3. Clean

Normalize case, whitespace, and symbols

→

4. Analyze

Search keywords and count terms

→

5.Validate

Check context and method limitations

→

6. Report

Present findings with ethical safeguards

Key principle: every transformation should be explainable, repeatable, and appropriate for the source material.

Python Text-Processing Examples

Illustrative code for introductory text analysis

```
text = "  Peace, Equality and Education!
"

clean_text = text.strip().lower()
words = clean_text.replace("!",
"").replace(",", "").split()

print(clean_text)
print(words)
print(words.count("equality"))
```

strip()

Removes surrounding whitespace.

lower()

Standardizes letters to lowercase.

replace() and split()

Removes selected punctuation and separates words.

The code is illustrative; actual findings require a defined and analyzed dataset.

What Can Text Analysis Reveal?

Potential outputs—not findings until the dataset is analyzed

Keyword frequency

Count terms such as education, peace, climate, and equality.

Document comparison

Compare topics across document types or publication periods.

Theme exploration

Group related terms for further human review.

Interpretation safeguard

A frequent word does not prove policy success, public opinion, or real-world impact. Results must be read in context and supported by suitable evidence.

Ethics Built into the Analytical Process

Core safeguards for responsible text analytics

Human dignity

Respect human rights and avoid unnecessary exposure of personal data.

Fairness

Check for bias across groups, cultures, and languages.

Privacy

Minimize personal data and follow applicable protections.

Transparency

Document sources, transformations, and assumptions.

Accountability

Make methods reviewable and correct errors where possible.

Context

Avoid drawing conclusions from keywords alone.

Connections to the UN Sustainable Development Goals

Illustrative links for organizing document themes

SDG 4 — Quality Education

Possible terms: education, access, learning

SDG 5 — Gender Equality

Possible terms: equality, empowerment

SDG 10 — Reduced Inequalities

Possible terms: inclusion, disparities

SDG 13 — Climate Action

Possible terms: climate, resilience

SDG 16 — Peace, Justice and Strong Institutions

Possible terms: justice, rights, accountability

SDG 17 — Partnerships for the Goals

Possible terms: cooperation, partnerships

Keyword references are a starting point, not a measurement of progress toward an SDG.

Limitations and Quality Assurance

What the analysis can—and cannot—claim

Results depend on document selection and language coverage.

Simple methods may miss context, ambiguity, and irony.

Keyword counts can overlook synonyms and related expressions.

Preprocessing can unintentionally change meaning.

Validate

Inspect sample outputs against original passages.

Reproduce

Record data sources, code, and settings.

Review

Use human judgment before making claims.

Conclusion

Python can make UN-related textual information easier to organize and explore—but responsible analysis requires ethical safeguards at every stage.

Technical methods help reveal patterns in text.

Ethics protects people and strengthens trust in analysis.

Human review is essential for meaningful interpretation.

Thank you • Questions and discussion

References

Foundational sources for the project

United Nations. (1945). Charter of the United Nations. https://www.un.org/en/about-us/un-charter

United Nations. (1948). Universal Declaration of Human Rights. https://www.un.org/en/about-us/universal-declaration-of-human-rights

United Nations. (n.d.). The 17 Sustainable Development Goals. https://sdgs.un.org/goals

Python Software Foundation. (n.d.). Python Documentation. https://docs.python.org/3/

pandas development team. (n.d.). pandas Documentation. https://pandas.pydata.org/docs/

Note: Include the specific UN documents analyzed and cite them in the final project submission.